

Constraint Encodings, Not Annealers, Dominate QUBO/HUBO Siting for Urban Heat-Island Interventions

The quhi project

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Abstract—We site parks, water bodies and cool pavement on synthetic 50 m city grids. Each instance is a constrained binary program, encoded as a QUBO or HUBO and scored on exact saturating cooling physics against a HiGHS-certified optimum. A slack-bit penalty QUBO with a corrected weight has the MILP optimum as its ground state on all 11 exhaustively enumerated instances. The second-order Taylor QUBO loses 10.3% of the achievable cooling at 300 m reach, where the cubic HUBO loses at most 0.02% and a least-squares-fitted QUBO at most 0.42%. The final annealing gap scales as $1/\Lambda$ with the penalty weight. Constraint-preserving Dicke-XY QAOA reaches a mean $P(\text{opt})$ of 0.43 at depth 6, against 1.00 for feasible-space simulated annealing. Path-integral Monte Carlo restricted to the feasible set passes a pre-committed comparability bar (benefit 0.981), but it made $3.6\times$ more proposals than feasible-space annealing; at equal proposals it scores 0.983 against 0.988. HiGHS certifies every hard-family instance in 0.77 s on average. Encodings and constraint handling, not the choice of annealer, decide the outcome. We observe no quantum or quantum-inspired advantage.

Index Terms—QUBO, HUBO, penalty methods, QAOA, simulated quantum annealing, urban heat island, benchmarking

I. INTRODUCTION

Urban planners who want to cool a city choose a small number of sites for parks, water and reflective pavement. The choice is a binary program with a budget, a one-option-per-cell rule and, often, an equity rule that every district receives something. Because such programs can be written as QUBOs, they are routinely offered as applications for quantum annealers and gate-model QAOA. Whether that is useful depends on how the constraints and the physics are encoded, and on what the quantum method is compared with.

We report a negative result with a precise cause. Using an audited open toolkit (quhi) and certified optima throughout, we find:

- the certified slack-penalty QUBO is exact (E1), but its penalty sets the analog gap, which scales as $1/\Lambda$ (E6);
- Taylor truncation to a QUBO misrepresents long-reach cooling (10.3% loss at 300 m), which a cubic HUBO or a fitted QUBO avoids (E2);
- constraint-preserving QAOA fixes feasibility but not quality: $P(\text{opt}) = 0.43$ at $p = 6$ against 1.00 for feasible-space SA (E8b);
- feasible-space path-integral Monte Carlo (FeasibleSQA) passes a pre-committed comparability bar, but at equal

proposals it trails feasible-space SA, 0.983 against 0.988 (E12, E12b);

- HiGHS certifies every hard-family instance in 0.77 s on average and a 16×16 LST-shaped tile in 1.67 s (E12b, E13).

The success criterion, fixed before the quantum experiments, is demanding by design: at matched wall-clock on certified instances, a method must meet the greedy planner and HiGHS MILP on both a park family and a mix + equity family. Matching simulated annealing is not enough.

II. PROBLEM

A. Cooling model

A city is a grid of 50 m cells [1]. Variable x_v , $v = (i, o)$, places option o (park, water body, cool pavement) on candidate cell i . Option o at cell i cools target cell k by $e_{ik} = A_o \exp(-d_{ik}/L_o)$; the defaults are 1.5 °C/100 m for parks, 2.0 °C/80 m for water and 0.6 °C/30 m for cool pavement, at 3, 5 and 1 cost units. Overlapping effects saturate:

$$\Delta T_k = \Delta T_{\max} \left[1 - \prod_v (1 - p_{kv} x_v) \right], \quad (1)$$

where $p_{kv} = \min(e_{kv}, \Delta T_{\max})/\Delta T_{\max}$ and $\Delta T_{\max} = 3^\circ\text{C}$. We minimise the exposure-weighted temperature $f(x) = \sum_k w_k (T_k^0 - \Delta T_k)$ with $w_k = 0.7 \text{ pop}_k / \sum \text{pop} + 0.3/N$. Truncating inclusion–exclusion at order K gives an additive model ($K=1$), a QUBO ($K=2$, a pessimistic bound by Bonferroni) or a cubic HUBO ($K=3$). An optional park-block synergy adds quartic terms.

B. Constraints and scoring

Constraints are an integer budget, at most one option per cell and optionally at least one intervention per district. Without equity, $-f$ is monotone submodular, so greedy enjoys a $(1 - 1/e)$ guarantee [2]. A plan’s $\text{benefit}_{\text{true}}$ is its exact-physics cooling divided by that of the MILP plan (infeasible = 0); $P(\text{opt})$ is the fraction of runs returning the certified optimum.

C. Why an audit came first

The earlier code base shipped energies of order 10^{10} and an all-zero SQA plan. The audit traced these to inequality constraints encoded as equalities in raw m^2 (coefficient ratio 1.72×10^9 against 2.0), an SQA action with β instead of β/P (single-qubit $\langle \sigma_z \rangle = -0.997$ against the exact -0.545), and

TABLE I
FINDINGS OF THE LEGACY AUDIT (F1–F7).

	finding	evidence (docs/AUDIT.md, results/audit.json)
F1	inequalities encoded as equalities, in raw m ²	$ Q _{\max} = 1.72 \times 10^9$ vs $ Q_{\text{phys}} _{\max} = 2.0$; infeasible global minimum
F2	shipped “results” are artefacts	energies are penalty magnitudes; SQA returns the all-zero plan
F3	SQA uses β instead of β/P	$\langle\sigma_z\rangle$: exact -0.545 , quhi -0.546 , legacy -0.997
F4	penalty theorem false for non-integer data	counter-example; corrected bound about $2\times$ smaller (median 2.06)
F5	automatic penalty weight not a bound	too small by up to a factor n ($Q = 1$, $n = 10$: 10 vs 100)
F6	QUBO $\leftrightarrow$ Ising round trip doubles couplings	round trip returns doubled off-diagonals
F7	wrong worked examples and “known optima”	claimed -19.25 where $f = -5$ and the optimum is -6

a penalty theorem that is false for non-integer data; 19 of its 163 tests fail. Table I lists the findings; none of the legacy numbers is used as evidence here. The legacy documents also appealed to asymptotic quantum-annealing results for disordered chains [3]; E6 shows why such results do not transfer to dense, penalty-dominated QUBOs.

III. METHODS

A. Exact penalty encoding

Constraints $a_k \cdot x$ ($\leq, \geq, =$) b_k have integer data. An equality contributes $(a \cdot x - b)^2$; a $\leq$ row contributes $(a \cdot x + \sum_j w_j s_j - b)^2$ with slack weights $w = (1, 2, \dots, 2^{m-1}, r)$ whose subset sums are exactly $\{0, \dots, b - \min a \cdot x\}$ (the *bounded* encoding; plain binary would admit slacks that are too large); a $\geq$ row uses negative slack; an at-most-one group contributes $\sum_{i < j} x_i x_j$. The sum $P(x, s)$ is integer and non-negative, at least 1 for every infeasible x , and 0 at the right slack for every feasible x . Hence, if f^* is the constrained optimum and $L \leq \min f$, any

$$\Lambda > f^* - L \quad (2)$$

makes every global minimiser of $f + \Lambda P$ feasible and optimal. We bound f^* from above with a repaired greedy point and call the result Λ_{safe} . The legacy statement, $\Lambda > f_{\max} - f_{\min}$ “because the penalty is at least Λ ”, needs integer data and is otherwise false (F4).

B. Alternatives to the safe penalty

Weak penalties use $\alpha \Lambda_{\text{safe}}$ with $\alpha \in \{0.01, 0.03, 0.1, 1\}$ followed by repair and feasible local search. *Unbalanced* and *linear* penalties need no slack bits but are exact only for suitable weights. *Rosenberg quadratisation* [4], [5] replaces a frequent pair uv by an auxiliary y with the penalty $M(uv - 2uy - 2vy + 3y)$; $M > S$, the sum of $|c_T|$ over terms that contain an auxiliary, is sufficient.

C. Solvers

- **Exact:** HiGHS MILP [6] on the constrained program; exhaustive enumeration and exact spectra for small models.

- **Classical heuristics:** SA [7] and Tabu [8] on penalty models (numba kernels with native k -local deltas); the cost-benefit greedy planner; *FeasibleSA*, Metropolis on feasible plans with add/remove and swap moves and no Λ ; *Tabu-on-F*, tabu search over the same neighbourhood.
- **Quantum-inspired:** SQA by path-integral Monte Carlo [9] with β/P on each slice; *FeasibleSQA*, the same PIMC with every Trotter slice feasible. Both are classical samplers, not models of annealing hardware [10].
- **Gate-model (simulated):** X-mixer QAOA [11] on penalty QUBOs (≤ 16 qubits) and *ConstrainedQAOA* with Dicke/XY mixers [12], simulated exactly on $\text{span}(\mathcal{F})$ from a dense eigendecomposition. It is a subspace simulation, not a compiled circuit.

D. Feasible-space samplers

FeasibleSA proposes, per sweep, n moves on a feasible plan: add or remove one bit, or swap an on-bit with an off-bit. A move that would leave $\mathcal{F}$ is rejected before f is evaluated; the rest follow Metropolis on the native objective with a geometric β schedule. FeasibleSQA applies the same moves to one of $P = 8$ Trotter slices at a time and samples

$$\pi(X) \propto \exp\left(-\frac{\beta}{P} \sum_k f(x^k) + J \sum_{k,i} s_i^k s_i^{k+1}\right), \quad (3)$$

with $J = \frac{1}{2} \ln \coth(\beta\Gamma/P)$, $s = 2x - 1$, periodic in k ; β and Γ are annealed jointly. It reproduces the exact single-qubit thermal $\langle\sigma_z\rangle$. Tabu-on-F searches the same add/remove and swap neighbourhood with aspiration and clears its tabu list when no admissible move remains.

E. Evaluation protocol for E12 and E12b

Each heuristic run receives $T^* = \max(\text{median } t_{\text{FSA}}, \min(t_{\text{HiGHS}}, 8\text{s}))$ of wall-clock as repeated batches of reads; a batch that would end after T^* is not started or is discarded, and a run longer than $1.5T^*$ scores 0. The best feasible plan is scored. A method is COMPARABLE on a family if $P(\text{feas}) = 1$, its mean $\text{benefit}_{\text{true}}$ is at least 0.95 of the best classical mean, and no run is over time; this rule was committed before E12 ran. E12b adds two pre-declared clocks: a tight clock (FeasibleSA’s own E12 wall-clock) and an equal-work budget W^* , the number of proposals FeasibleSQA used in E12, given to every method.

All runs are seeded; the environment (Python 3.11, numpy 2.4, scipy 1.17 with HiGHS, numba 0.67, single-threaded solvers) is recorded in `results/env.json`.

IV. RESULTS

A. Encodings and surrogates (E1, E2, E5, E11)

Exhaustive enumeration of decision and slack bits shows the penalty QUBO ground state equal to the MILP optimum on 11 of 11 instances (Table IV); Λ_{safe} is $1.81\text{--}2.21\times$ smaller than the textbook range bound. Surrogate order matters once cooling reach is long (Table V, Fig. 1): at $L = 300\text{m}$ the

TABLE II
E5: NATIVE QUARTIC HUBO VS ROSENBERG QUBO (3 CERTIFIED INSTANCES).

formulation	solver	vars	P(opt) raw	pp	benefit
QUBO (Rosenberg)	Greedy	192	0.000	1.000	0.849
QUBO (Rosenberg)	SA	192	0.000	1.000	0.837
QUBO (Rosenberg)	SQA	192	0.000	0.997	0.930
QUBO (Rosenberg)	Tabu	192	0.000	1.000	0.847
native HUBO	Greedy	30	0.003	0.976	0.921
native HUBO	SA	30	0.000	0.997	0.920
native HUBO	SQA	30	0.000	0.993	0.960
native HUBO	Tabu	30	0.990	1.000	0.999

TABLE III
E11: NATIVE HUBO VS ROSENBERG QUBO SPECTRA (QUICK RUN, 2 INSTANCES).

model	qubits	final gap	P(ground), $T=10$
A: native cubic HUBO + slack penalty	9	$9.4 \times 10^{-5} / 1.1 \times 10^{-5}$	3.1×10^{-3}
B: Rosenberg QUBO of A	16	$9.4 \times 10^{-5} / 1.1 \times 10^{-5}$	2.4×10^{-5}
C: native HUBO on $\mathcal{F}$ (diagonal only)	6	$2.4 \times 10^{-2} / 3.7 \times 10^{-3}$	$4.5 \times 10^{-2} / 4.1 \times 10^{-2}$

TABLE IV
E1: PENALTY-QUBO GROUND STATE VS MILP OPTIMUM (EXHAUSTIVE).

instance	dec.	slack	MILP opt. ($^{\circ}\text{C}$)	QUBO ground ($^{\circ}\text{C}$)	match	Λ_{safe}	range/safe
6x6_s0_park	5	3	30.818	30.818	yes	1.906	1.881
6x6_s1_park	5	3	30.755	30.755	yes	1.825	1.839
6x6_s2_park	5	3	30.720	30.720	yes	1.845	1.854
6x6_s3_park	5	3	30.823	30.823	yes	2.045	1.815
8x8_s0_park	12	4	30.584	30.584	yes	2.961	2.124
8x8_s1_park	12	4	30.507	30.507	yes	2.330	2.208
8x8_s2_park	12	4	30.738	30.738	yes	2.417	2.193
8x8_s3_park	12	4	30.757	30.757	yes	2.706	2.146
6x6_s0_mix	15	3	30.369	30.369	yes	4.850	2.063
6x6_s1_mix	15	3	30.339	30.339	yes	4.538	2.018
8x8_s1_park + equity	12	11	30.552	30.552	yes	2.403	2.142

TABLE V
E2: SHARE OF ACHIEVABLE COOLING LOST BY EACH SURROGATE'S OPTIMUM (EXACT PHYSICS).

L (m)	additive $K=1$	QUBO $K=2$ (Taylor)	QUBO (LSQ fit)	HUBO $K=3$
50	0.20\{\}%	0.00\{\}%	0.00\{\}%	0.00\{\}%
100	0.00\{\}%	0.20\{\}%	0.02\{\}%	0.00\{\}%
200	0.29\{\}%	2.81\{\}%	0.21\{\}%	0.02\{\}%
300	0.00\{\}%	10.30\{\}%	0.42\{\}%	0.00\{\}%

Taylor QUBO loses 10.30% of the achievable cooling, the cubic HUBO none, and a least-squares QUBO with the same 253 coefficients 0.42%. Higher order should stay native: Tabu solves a 30-variable quartic HUBO on 99.0% of reads but on 0% after Rosenberg quadratisation to 187–197 variables (E5, Table II). Quadratising a cubic HUBO leaves the slack-penalty final gap unchanged, adds 7 qubits and lowers short-anneal ground-state probability about $100\times$ (E11, quick run, Table III); removing the penalty enlarges the gap about $250\times$.

B. Classical baselines on penalty models (E3, E7, E9, E10)

Table VI gives the median raw success probability on the safe-penalty QUBOs. Tabu solves single-intervention instances up to 27 variables on every read (0.66 at 46), whereas no sampler reaches the optimum on raw reads of the mix + equity family (42–89 variables). SQA beats SA at equal sweeps, but SA given as many flip attempts ("SA-matched") is comparable on parks, and SQA's +0.02–0.03 benefit on mix + equity

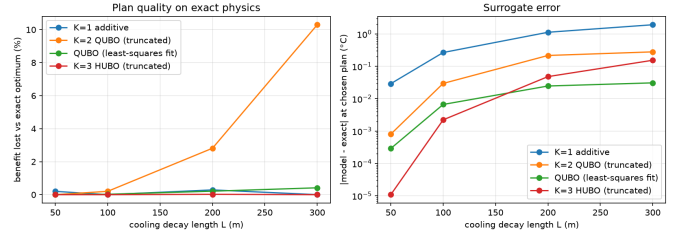


Fig. 1. E2 surrogate fidelity versus decay length (results/E2_fidelity/fidelity.png).

TABLE VI
E3: MEDIAN RAW $P(\text{opt})$ PER READ ON SAFE-PENALTY QUBOS.

family / vars	Greedy	SA	SA-matched	SQA	SQA-fixedT	Tabu	QAOA($p=4$)
park / 13	0.20	0.48	1.00	1.00	1.00	1.00	0.01
park / 17	0.06	0.03	0.48	0.56	0.84	1.00	—
park / 27	0.00	0.00	0.00	0.00	0.00	1.00	—
park / 32	0.00	0.00	0.00	0.00	0.00	0.94	—
park / 46	0.00	0.00	0.00	0.00	0.00	0.66	—
mix+equity / 42	0.00	0.00	0.00	0.00	0.00	0.00	—
mix+equity / 56	0.00	0.00	0.00	0.00	0.00	0.00	—
mix+equity / 89	0.00	0.00	0.00	0.00	0.00	0.00	—

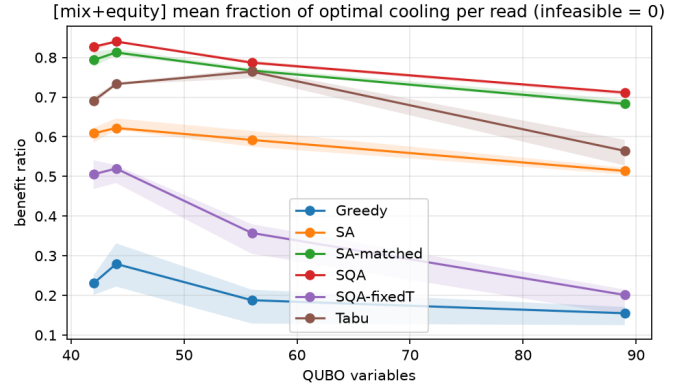


Fig. 2. E3: mean benefit ratio per read versus size on the mix + equity family (results/E3_scaling/benefit_vs_n_mix_equity.png).

costs about $2.4\times$ the wall time (Fig. 2). The greedy planner is optimal on 8 of 15 park instances and infeasible on 5 of 9 mix instances. On the E7 showcase (14×14 , 81 decision variables, Fig. 3) MILP certifies in 10.8 s and the greedy planner returns the same plan, lowering the exposure temperature from 32.311 to 31.141 $^{\circ}\text{C}$.

Weak penalties plus repair are the best penalty-based recipe (E9, Table VII): at $\alpha = 0.01$, SA-matched reaches post-processed $P(\text{opt}) = 0.80$ and SQA 0.77, against at most 0.18 at safe Λ ; FeasibleSA reaches 0.46 per read without any penalty, and MILP certifies in 3.8 s on average. Slack-free penalties (E10, quick run, Table VIII) cut the dynamic range from 163 to 16.5 at $0.1 \Lambda_{\text{safe}}$ with an optimal ground state on these two instances, but are not certified in general.

C. The penalty sets the analog gap (E4, E6)

On a 14-qubit instance the minimum gap of $H(s) = -(1-s) \sum X + sH_P$ sits at $s = 1$ and scales as $1/\Lambda$: $\Delta(s=1) = 7.1 \times 10^{-5}$ at $0.1 \Lambda_{\text{safe}}$ and 7.1×10^{-7} at $10 \Lambda_{\text{safe}}$

Baseline -> optimised intervention plans (14x14, 3 intervention types, equity)

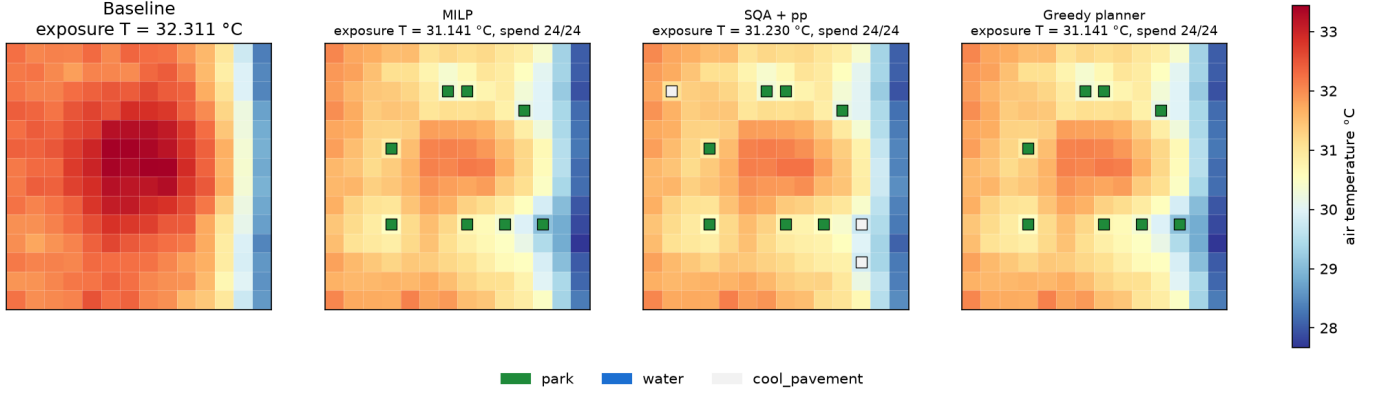


Fig. 3. E7 showcase: baseline to optimised plans on a 14×14 city with three intervention types and equity (results/E7_showcase/transition.png).

TABLE VII

E9: WEAK PENALTY ($\alpha\Lambda_{\text{safe}}$) PLUS REPAIR, MIX + EQUITY, 9 INSTANCES.

solver	α	P(feas)	P(opt) raw	P(opt) pp	benefit pp
FeasibleSA	—	1.000	0.460	—	—
Greedy planner	—	0.444	0.444	—	—
MILP (HiGHS)	—	1.000	1.000	—	—
SA	0.01	0.608	0.017	0.503	0.941
SA	1	0.979	0.000	0.031	0.813
SA-matched	0.01	0.715	0.168	0.804	0.976
SA-matched	1	1.000	0.003	0.184	0.896
SQA	0.01	0.833	0.156	0.767	0.979
SQA	1	1.000	0.003	0.108	0.880
Tabu	0.01	0.792	0.078	0.321	0.903
Tabu	1	0.983	0.000	0.009	0.748

TABLE VIII

E10: PENALTY FORMS (QUICK RUN, 2 INSTANCES).

form	weight	vars	dyn. range	ground feas.	ground opt.	P(opt) pp
quadratic slack (exact)	Λ_{safe}	13	163.0	1.000	1.000	0.719
unbalanced	$0.1 \cdot \Lambda_{\text{safe}}$	10	16.5	1.000	1.000	0.859
unbalanced	$0.3 \cdot \Lambda_{\text{safe}}$	10	49.0	1.000	1.000	0.766
unbalanced	$1.0 \cdot \Lambda_{\text{safe}}$	10	163.0	1.000	1.000	0.812
linear	$0.5 \cdot \mu$	10	0.6	0.000	0.000	1.000
linear	$1.0 \cdot \mu$	10	0.4	0.500	0.500	0.500
linear	$2.0 \cdot \mu$	10	1.0	1.000	0.000	1.000

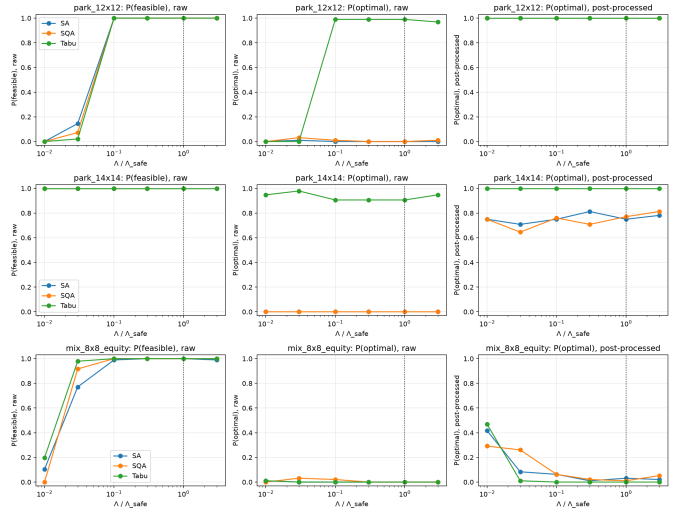


Fig. 4. E4: raw and post-processed feasibility and success versus $\Lambda/\Lambda_{\text{safe}}$ (results/E4_penalty/penalty_sweep.png).

TABLE IX

E6: 14-QUBIT SPECTRA AND QAOA VERSUS PENALTY WEIGHT.

$\Lambda/\Lambda_{\text{safe}}$	ground = opt.	$\Delta(s=1)$	$1/\Delta^2$	QAOA $P(\text{opt}), p=6$
0.1	yes	7.1×10^{-5}	2.0×10^8	0.0019
0.3	yes	2.4×10^{-5}	1.8×10^9	0.0018
1	yes	7.1×10^{-6}	2.0×10^{10}	0.0017
3	yes	2.4×10^{-6}	1.8×10^{11}	0.0017
10	yes	7.1×10^{-7}	2.0×10^{12}	0.0017

(Table IX, Fig. 5). The adiabatic scale $1/\Delta^2$ [13] therefore grows from 2.0×10^8 to 2.0×10^{12} . X-mixer QAOA at $p = 6$ places 0.17–0.19% of its probability on the optimum while reporting an approximation ratio of about 0.998, a number that mostly measures “feasible and low penalty”. Weak penalties help samplers but give up certification: $0.01 \Lambda_{\text{safe}}$ plus repair lifts post-processed $P(\text{opt})$ from at most 0.03 to 0.42–0.47 on a mix + equity instance (E4, Fig. 4).

D. Constraint-preserving QAOA (E8, E8b)

Dicke/XY mixers keep the state feasible and raise $P(\text{opt})$ 10–70 $\times$ over X-mixer penalty QAOA at equal depth (E8). They do not reach the classical bar. Table X and Fig. 6 show the depth curve on three “exactly 3 parks” cities ($|\mathcal{F}| = 120$): Dicke-XY complete rises from 0.134 at $p = 1$ to 0.427 at $p = 6$ (maximum 0.65), while FeasibleSA returns the optimum

in every run in 0.21 s. Mixer connectivity matters more than depth: ring-XY at $p = 6$ (0.111) is below complete-XY at $p = 1$. On mix + equity, XY-QAOA at $p = 2$ reaches $P(\text{opt}) = 0.003$ (benefit 0.61) against FeasibleSA’s 0.39 (benefit 0.92).

E. Feasible-space PIMC and its ablation (E12, E12b)

Family H has nine mix + equity instances on 8×8 to 12×12 grids with a budget equal to the grid side. Under a matched wall-clock T^* per run (1.4–20 s), FeasibleSQA

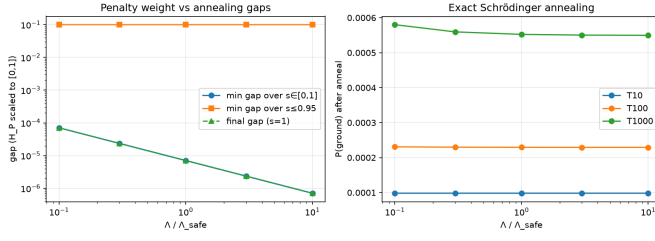


Fig. 5. E6: final gap and adiabatic scale versus penalty weight (14 qubits).

TABLE X

E8B: EXACT $P(\text{opt})$ VS QAOA DEPTH, MEAN [MIN–MAX] OVER 3 CITIES.

method	$p=1$	$p=2$	$p=3$	$p=4$	$p=6$
Dicke-XY complete	0.134	0.242 [0.08–0.37]	0.309	0.376	0.427 [0.17–0.65]
Dicke-XY ring	0.034	0.067 [0.02–0.14]	0.057	0.066	0.111 [0.02–0.26]
X-mixer penalty QAOA	0.004	0.005	0.006	0.008	0.009

method	$P(\text{opt})$	benefit	wall (s)
MILP (HiGHS)	1.000	1.000	0.317
Greedy planner	0.667	1.000	8.54×10^{-4}
uniform on F	0.008	0.849	0.000
FeasibleSA	1.000	1.000	0.206

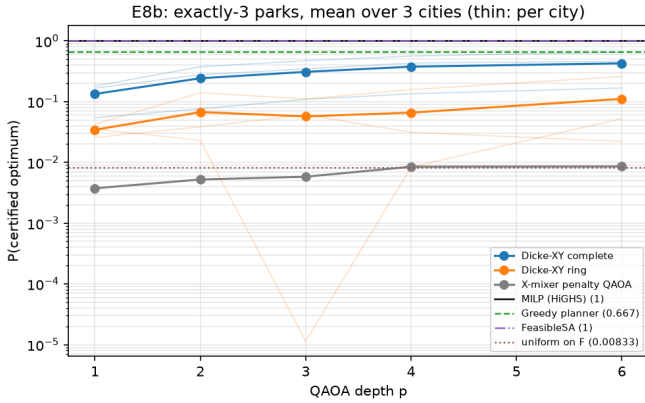


Fig. 6. E8b: $P(\text{opt})$ versus depth p for Dicke-XY complete and ring mixers and the X-mixer penalty control.

reaches a mean $\text{benefit}_{\text{true}}$ of 0.981 and passes the pre-committed rule (Table XI). So do FeasibleSA (0.973) and Tabu-on-F (0.976); the greedy planner is infeasible on 6 of 9 instances; ConstrainedQAOA (0.658 at $p=2$, 0.665 at $p=4$) fits only the 8×8 cities, and MILP reaches 1.000 in 0.81 s.

The ablation explains the ordering. HiGHS never set T^* , so the clock was not padded; instead, FeasibleSQA made $3.6\times$ as many proposals in the same time (4.66 M against 1.30 M per run), because moves that would leave $\mathcal{F}$ are rejected before f is evaluated. At an equal budget of 4.53 M proposals per run, FeasibleSA scores 0.988 and FeasibleSQA 0.983 (ahead on 1 instance, tied on 6, behind on 2; Table XII). FeasibleSQA is within 5% of FeasibleSA; it is not ahead. FeasibleSQA needs less wall-clock per proposal (7.1 s against 25.0 s for the same count), an implementation property of early rejection, not a quantum effect. Tabu-on-F does not improve with more work

(0.977 on both clocks), and QAOA-seeded FeasibleSA still loses the $|\mathcal{F}| = 2484$ city on the tight clock because the subspace simulation alone exceeds it. Fig. 7 places every E12 method on the benefit–time plane.

F. An LST-shaped tile (E13)

A 16×16 tile with a compact hot core, population on the core and only 2 candidate lots left in the core ($n = 108$, $|\mathcal{F}| > 4000$) did not create a classical gap (Table XIII). HiGHS certifies it in 1.67 s; greedy fails on equity; the weak-penalty hybrid finds the optimum on both seeds. FeasibleSQA (0.9998) above FeasibleSA (0.9569) on two tight-clock seeds is not evidence of an advantage, for the reason E12b gives. Fig. 8 shows the tile and the returned plans; Tabu-on-F falls to 0.37 on one seed.

V. DISCUSSION

Criterion not met. Gate-model QAOA misses the bar with penalties (it learns the constraint, not the objective) and with constraint-preserving mixers (it stays well below FeasibleSA and MILP at every simulated depth). Feasible-space PIMC only ties feasible-space SA.

Encodings decide. Every improvement to a quantum or quantum-inspired method in this study was an encoding change: certified but gentler penalties, native higher-order terms, and moves or mixers that respect the constraints. Each helped the classical methods at least as much.

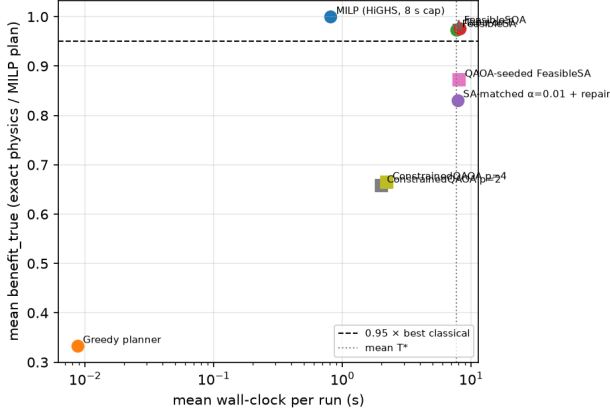
Hold-ups, in order of impact. (i) Slack penalties compress the objective, so the analog gap scales as $1/\Lambda$ (E6, E11). (ii) Quadratisation multiplies variables $6.4\times$ and blocks single-flip dynamics (E5). (iii) At equal work, PIMC advantages over SA vanish (E3, E9, E12b). (iv) The objective is submodular, so greedy is strong wherever equity does not bind. (v) Constraint-preserving mixers fix feasibility but not optimality at simulable depths (E8b). (vi) QAOA approximation ratios near 1 are cosmetic on penalty-dominated scales (E6). (vii) No constrained model was run on a QPU or hybrid solver.

What hardware could test. A fair test would send the constrained model natively, as a CQM to a hybrid solver, on the E8b “exactly 3” city and a park-only control city. A safe- Λ slack QUBO should not be sent to a QPU, because its gap is set by Λ . No D-Wave result exists in this work; the repository hook is inactive without an API token.

VI. LIMITATIONS

Cities and the E13 tile are synthetic, with illustrative intervention parameters; the saturation law is a surrogate for CFD, and MILP certifies the $K=2$ surrogate. SQA and FeasibleSQA are PIMC, not hardware quantum annealing [10]. ConstrainedQAOA is a subspace simulation on $\text{span}(\mathcal{F})$, not a compiled circuit. E12/E12b use three instances and two seeds per cell, and E10/E11 are quick runs on two instances.

E12 family H (o classical, * quantum-inspired, s quantum sim)



E12 family C (o classical, * quantum-inspired, s quantum sim)

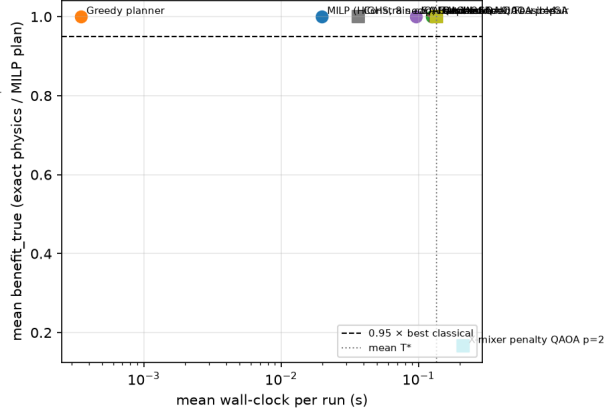


Fig. 7. E12: mean benefit_{true} versus mean wall-clock per run for family H (left) and the park control C (right); the dashed line is 0.95 of the best classical mean (results/E12_comparable/benefit_vs_time.png).

TABLE XI

E12 FAMILY H AT MATCHED WALL-CLOCK T^* .

method (family H)	P(feas)	P(opt)	benefit	min	wall (s)	comp.
MILP (HiGHS, 8 s cap)	1.000	1.000	1.000	1.000	0.809	yes
Greedy planner	0.333	0.333	0.333	0.000	0.009	no
FeasibleSA	1.000	0.778	0.973	0.813	7.693	yes
Tabu-on-F	1.000	0.778	0.976	0.708	8.120	yes
SA-matched $\alpha=0.01$ + repair	0.833	0.722	0.831	0.000	7.847	no
FeasibleSQA	1.000	0.722	0.981	0.854	7.975	yes
QAOA-seeded FeasibleSA	1.000	0.833	0.873	0.000	8.095	no
ConstrainedQAOA p=2	1.000	0.500	0.658	0.000	2.008	no
ConstrainedQAOA p=4	1.000	0.167	0.665	0.000	2.221	no

TABLE XII

E12B FAMILY H AT EQUAL PROPOSALS W^* .

method	clock	P(opt)	benefit	min	wall (s)	prop. (M)	w/t/l
MILP (HiGHS, 8 s cap)	none	1.000	1.000	1.000	0.77	—	—
FeasibleSA	equal_work	0.889	0.988	0.854	24.99	4.53	—
Tabu-on-F	equal_work	0.833	0.977	0.708	11.32	4.65	1/6/2
FeasibleSQA	equal_work	0.778	0.983	0.813	7.09	4.53	1/6/2
QAOA-seeded FeasibleSA	equal_work	1.000	1.000	1.000	2.28	2.20	0/3/0

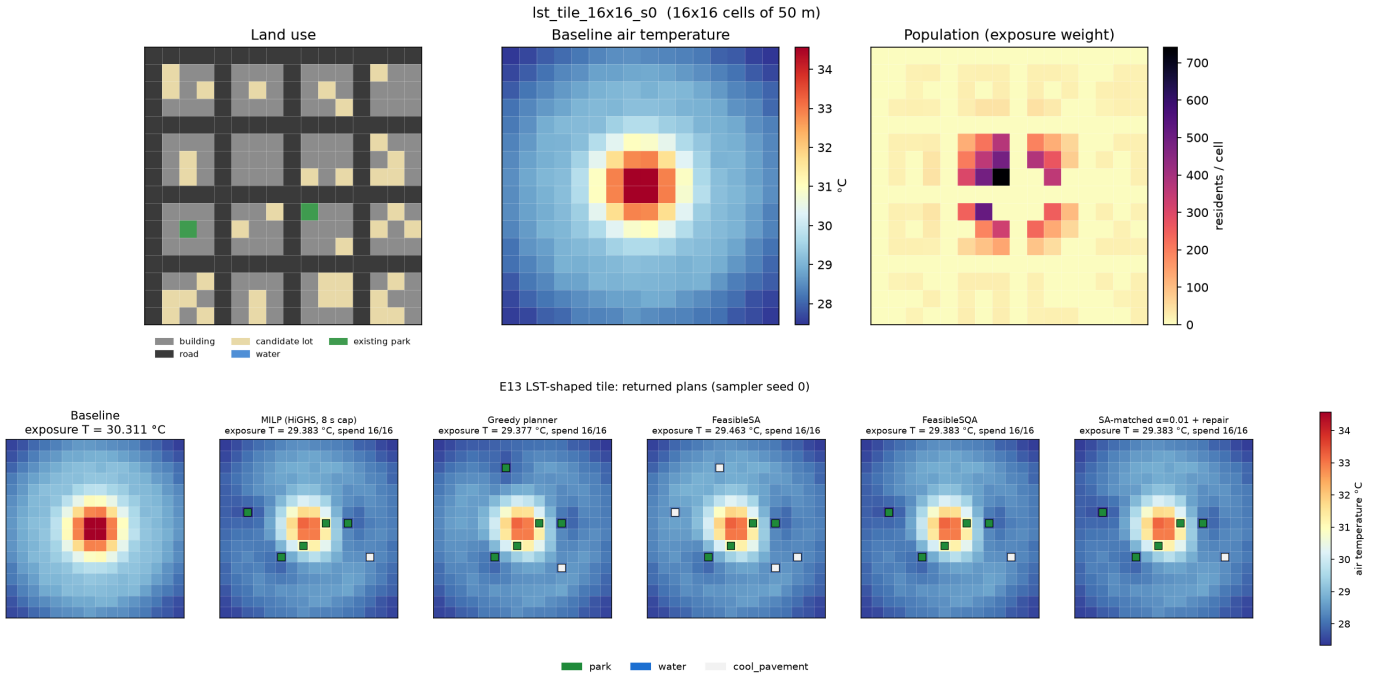


Fig. 8. E13: the LST-shaped tile (land use, baseline temperature, population) and the plans returned by MILP, greedy, FeasibleSA, FeasibleSQA and the weak-penalty hybrid (sampler seed 0).

REPRODUCIBILITY

Every table is generated from the committed CSVs by `latex/tools/make_tables.py`; experiments

rerun with `python scripts/run_experiments.py --only Ek`, and the legacy audit with `python`

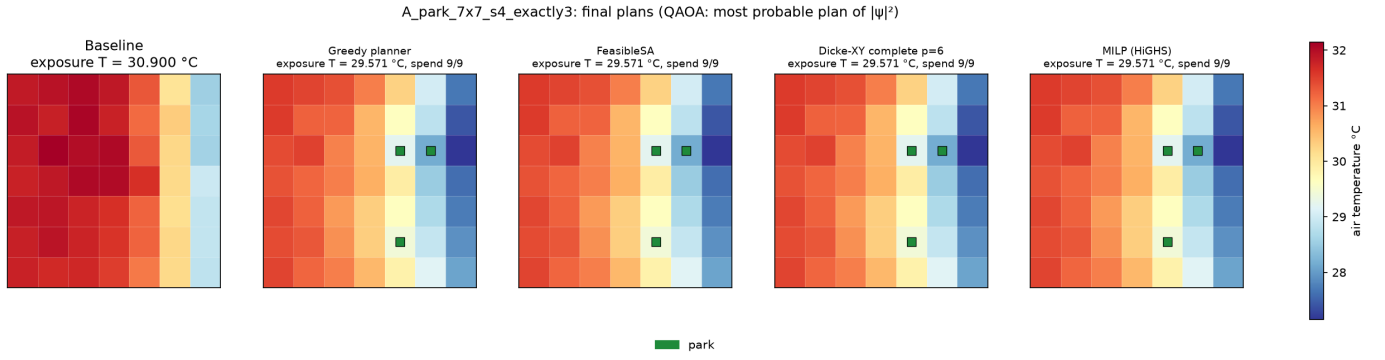


Fig. 9. Final plans on the E8b “exactly 3 parks” city: greedy, FeasibleSA, Dicke-XY $p = 6$ (most probable plan) and MILP, on one locked temperature scale. Animated versions are supplementary: `results/ANIM/compare_temp_A_park_7x7_s4_exactly3.gif` and `results/ANIM/compare_temp_B_mix_10x10_s0.gif`.

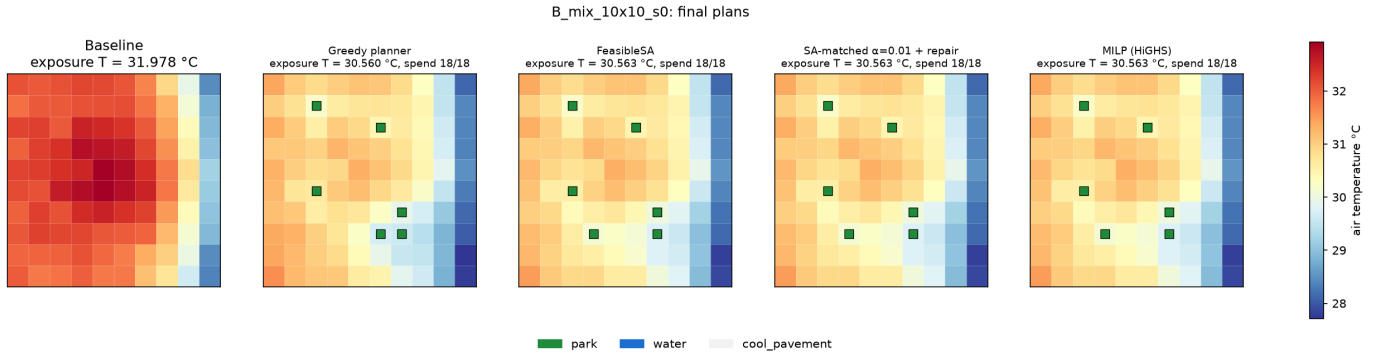


Fig. 10. Final plans on the E9 10×10 mix + equity city: greedy (infeasible on equity), FeasibleSA, the weak-penalty hybrid (in place of QAOA, since $|\mathcal{F}| > 4000$) and MILP.

TABLE XIII
E13 LST-SHAPED TILE, TIGHT CLOCK 68.59 s, TWO SEEDS.

method	$P(\text{feas})$	$P(\text{opt})$	benefit	wall (s)
MILP (HiGHS, 8 s cap)	1.0	1.0	1.0000	1.67
Greedy planner	0.0	0.0	0.0000	0.06
FeasibleSA	1.0	0.5	0.9569	66.56
Tabu-on-F	1.0	0.0	0.6412	67.35
FeasibleSQA	1.0	0.5	0.9998	63.72
SA-matched $\alpha=0.01$ + repair	1.0	1.0	1.0000	64.68

`scripts/audit_legacy.py`. Animated comparisons are in `results/ANIM/`.

ACKNOWLEDGEMENTS

The experiments, the toolkit and this manuscript were produced in the `s-siyanwal/q_uhi` repository (branch `claude/clever-gates-ezote4`, commit `e649582`). No external data, cloud solver or quantum hardware was used.

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